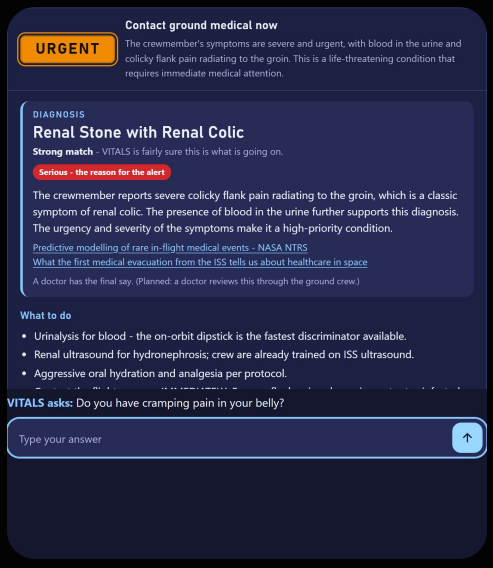
VITALS

When the doctor is 20 minutes away, the crew still needs an answer now.

VITALS is an onboard medical assistant a non-medical crewmember can talk to in plain words. It runs with no internet, knows spaceflight medicine, puts urgency first — and shows the source behind every answer.

The problem

On Mars, a message to the flight surgeon takes 4 to 40 minutes each way, with blackouts of up to two weeks. Most crews will have no physician on board.



Top features

- Offline — the AI model is built in; no data leaves the habitat
- Sourced — 35 spaceflight conditions from NASA and peer-reviewed research; the AI can't invent a citation
- Safe — a rules engine can raise the alarm even when the AI doesn't
- Honest — no made-up vital signs, no fake percentages
- Fixable — a "Wrong?" button on everything it understood

Made with GAMMA

How it works

1. Ask — the crewmember types how they feel
2. Read — pull the symptoms out of the words
3. Retrieve — pick the cited research pages that fit
4. Reason — a local AI reads only those pages
5. Check — drop anything invented; rules engine can always escalate
6. Answer — urgency first, then what it could be, why, and what to do

Works now

astronaut screen, reasoning over 35 conditions, safety backstop, every answer saved as a case.

Planned

ground crew screen, doctor screen, sensors (heart rate, blood oxygen, blood pressure, temperature), Jetson + three Raspberry Pi screens, voice input.

Technical specs

local language model (llama3.2, 3B) via Ollama · deterministic rules engine · Python · Docker · NVIDIA Jetson Orin Nano 8 GB (planned) · 262 automated tests · 18 of 19 urgent complaints escalated (Oct 17)

Team TETHER

Joshua Collado · Joaquin · Cruz

South Texas ISD Science Academy · Mercedes, Texas

Teacher: Ms. Sylvia Odoms · NASA HUNCH 2026–27

NASA project: Medical Diagnostic System with Machine Learning & Artificial Intelligence (From a non-Medical Person's Perspective)

Website

Technical specs

Made with GAMMA

A student prototype for diagnosis and decision support — a doctor has the final say.